

# Nils Froleyks

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## RESEARCH INTERESTS

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Hardware Verification, Automated Reasoning, Symbolic AI and Planning.  
Building theoretical foundations and designing scalable algorithms to enable trustworthy chip designs.

## ACADEMIC EXPERIENCE

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| <b>Postdoctoral Researcher</b><br>KU Leuven, Belgium<br>with <b>Prof. Bart Bogaerts</b> in the Declarative Languages and Artificial Intelligence section | October 2025 – Present   |
| <b>University Assistant</b><br>Johannes Kepler University Linz, Austria<br>Teaching duties included courses on Model Checking, Logic, and Debugging.     | March 2020 – August 2025 |

## EDUCATION

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| <b>Johannes Kepler University Linz</b> , Linz, Austria<br>Ph.D in Computer Science with Distinction<br>Thesis: Deep Integration of SAT Solving and Model Checking<br>Supervisors: Prof. Armin Biere, Prof. Martina Seidl | March 2020 – August 2025                |
| <b>Karlsruhe Institute of Technology</b> , Karlsruhe, Germany<br>M.Sc in Computer Science<br>Thesis: PASAR – Planning as Satisfiability with Abstraction Refinement<br>Supervisors: Prof. Peter Sanders                  | 2018 – 2020<br><b>Best Thesis Award</b> |
| <b>Karlsruhe Institute of Technology</b> , Karlsruhe, Germany<br>B.Sc in Computer Science<br>Thesis: Using an Algorithm Portfolio to Solve Sokoban<br>Supervisors: Prof. Peter Sanders                                   | 2013 – 2017                             |

## INDUSTRIAL COLLABORATION

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| <b>Intel Corporation</b><br>Contributed to certification for hardware model checking. Monthly meetings and joint workshops. | Nov 2023 – Present |
| <b>Woodpecker Technologies</b> , Singapore<br>External consultant. Pre-silicon verification using symbolic formal methods.  | 2022 – 2023        |

## CAREER BREAKS

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| <b>Parental Leave</b>                                                                                | February – November 2022 |
| <b>CAV Distinguished Paper Award</b><br>37th International Conference on Computer Aided Verification | 2025                     |
| <b>Third Place in Combinatorial Reconfiguration Challenge</b><br>CoRe Challenge 2022, Virtual        | 2022                     |
| <b>Best Master’s Thesis</b><br>Karlsruhe Institute of Technology, Germany                            | 2020                     |
| <b>First Prize (Partial Gold Medal)</b><br>The Sparkle Planning Challenge, ICAPS, Berkeley, USA      | 2019                     |

## SCIENTIFIC EVENT ORGANIZATION

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| <b>Organizer, Hardware Model Checking Competition (HWMCC) 2025</b> | FMCAD, Menlo Park, USA        |
| <b>Organizer, Hardware Model Checking Competition (HWMCC) 2024</b> | FMCAD, Prague, Czech Republic |
| <b>Organizer, SAT Competition 2021</b>                             | SAT Conference, Barcelona     |

**Organizer, Hardware Model Checking Competition (HWMCC) 2020**  
**Organizer, SAT Competition 2020**

FMCAD, Virtual  
 SAT Conference, Alghero, Italy

## SKILLS

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**Programming Languages:** C++, Python

**Languages:** German (native), English (fluent)

## TEACHING & SUPERVISION

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**Guest Lecturer in SAT solving**

**Spring Semester 2026**

KU Leuven, Belgium

Advanced course for master's students in computer science.

**Supervised Bachelor's Thesis**

**2024**

Johannes Kepler University Linz, Austria

Ternary simulation for IC3 Model Checking.

**Lecturer in Debugging**

**Summer Semester 2024**

Johannes Kepler University Linz, Austria

Advanced course for master's students in computer science and artificial intelligence.

**Lecturer in Model Checking**

**Winter Semester 2024**

Johannes Kepler University Linz, Austria

Advanced course for master's students in computer science and artificial intelligence.

**Teaching Assistant in Computer Science and Artificial Intelligence Courses**

**2020–2024**

Johannes Kepler University Linz, Austria

At both the Master's and Bachelor's levels, hands-on teaching courses include Model Checking, Formal Models, Logic, SAT Solving, Practical Work in AI, and Computational Logics for AI.

**Teaching Assistant in Algorithms for Planar Graphs**

**2018**

Karlsruhe Institute of Technology, Germany

**Teaching Assistant in Theoretical Foundations of Informatics**

**2015–2019**

Karlsruhe Institute of Technology, Germany

## SOFTWARE

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**Certifaiger:** a certificate checker for the hardware model checking certificates

<https://github.com/Froleyks/certifaiger>

**Voirraig:** a certifying model checker

<https://github.com/Froleyks/voirraig>

**Cerbtora:** a word-level certificate checker for hardware model checking

<https://github.com/Froleyks/cerbtora>

**CaDiCaL:** an award-winning state-of-the-art SAT solver

<https://github.com/arminbiere/cadical>

## CONTACTS FOR REFERENCES

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**Prof. Bart Bogaerts**, KU Leuven, Belgium. [bart.bogaerts@kuleuven.be](mailto:bart.bogaerts@kuleuven.be)

**Prof. Armin Biere**, University of Freiburg, Germany. [armin.biere@gmail.com](mailto:armin.biere@gmail.com)

**Prof. Martina Seidl**, Johannes Kepler University Linz, Austria. [martina.seidl@jku.at](mailto:martina.seidl@jku.at)

## PUBLICATIONS

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- [1] Florian Pollitt Mathias Fleury, Katalin Fazekas, **Nils Froleyks**, and Armin Biere. *CaDiCaL 3.0*. Submitted to SAT'26
- [2] **Nils Froleyks**, Emily Yu, Bart Bogaerts, Armin Biere, and Keijo Heljanko. *Liveness Proofs for Hardware Model Checking*. Submitted to CAV'26
- [3] **Nils Froleyks**, Emily Yu. *Hardware Model Checking Certification with Certifaiger and Cerbtora*. Submitted to IJCAR'26
- [4] Armin Biere, Katalin Fazekas, Mathias Fleury, **Nils Froleyks**, and Florian Pollitt. *Clausal Equivalence Checking*. Journal Submission under review

- [5] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying Constraints in Hardware Model Checking*. International Symposium on Formal Methods (FM), 2026
- [6] **Nils Froleys**, Emily Yu, Mathias Preiner, Armin Biere, and Keijo Heljanko. *Introducing Certificates to the Hardware Model Checking Competition*. International Conference on Computer Aided Verification (CAV), 2025
- [7] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL 2.0*. International Conference on Computer Aided Verification (CAV), 2024
- [8] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal equivalence sweeping*. Formal Methods in Computer Aided Design (FMCAD), 2024
- [9] Armin Biere, **Nils Froleys**, and Mathias Preiner. *Hardware Model Checking Competition 2024*. Formal Methods in Computer Aided Design (FMCAD), 2024
- [10] Armin Biere, Katalin Fazekas, Mathias Fleury, and **Nils Froleys**. *Clausal congruence closure*. Theory and Applications of Satisfiability Testing (SAT), 2024
- [11] **Nils Froleys**, Emily Yu, Armin Biere, and Keijo Heljanko. *Certifying phase abstraction*. International Joint Conference on Automated Reasoning (IJCAR), 2024
- [12] **Nils Froleys**, Emily Yu, and Armin Biere. *Ternary Simulation as Abstract Interpretation (Work in Progress)*. Methoden und Beschreibungssprachen zur Modellierung und Verifikation von Schaltungen und Systemen (MBMV), 2024
- [13] Armin Biere, **Nils Froleys**, and Wenxi Wang. *Cadiback: Extracting backbones with cadical*. Theory and Applications of Satisfiability Testing (SAT), 2023
- [14] Armin Biere, Mathias Fleury, **Nils Froleys**, and Marijn JH Heule. *The SAT Museum*. Pragmatics of SAT, 2023
- [15] **Nils Froleys**, Emily Yu, and Armin Biere. *BIG Backbones*. Formal Methods in Computer-Aided Design (FMCAD), 2023
- [16] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Towards compositional hardware model checking certification*. Formal Methods in Computer-Aided Design (FMCAD), 2023
- [17] Emily Yu, **Nils Froleys**, Armin Biere, and Keijo Heljanko. *Stratified certification for k-induction*. Formal Methods in Computer-Aided Design (FMCAD), 2022
- [18] **Nils Froleys**, and Armin Biere. *Single clause assumption without activation literals to speed-up IC3*. Formal Methods in Computer-Aided Design (FMCAD), 2021
- [19] Tomás Balyo, and **Nils Froleys**. *Ai assisted design of sokoban puzzles using automated planning*. International Conference on ArtsIT, Interactivity and Game Creation, 2021
- [20] **Nils Froleys**, Marijn Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *SAT competition 2020*. Journal of Artificial Intelligence, 2021
- [21] **Nils Froleys**, Tomas Balyo, and Dominik Schreiber. *PASAR—Planning as Satisfiability with Abstraction Refinement*. International Symposium on Combinatorial Search, 2019
- [22] **Nils Froleys**, and Tomás Balyo. *Using an algorithm portfolio to solve Sokoban*. International Symposium on Combinatorial Search, 2017

## NON-PEER-REVIEWED PUBLICATIONS

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- [1] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, Florian Pollitt, and André Schidler. *CaDiCaL, IsaSAT and Kissat Entering the SAT Competition 2025*. SAT Competition 2025.
- [2] **Nils Froleys**, Emily Yu, and Armin Biere. *Challenging Certificates from Model Checking*. SAT Competition 2025.
- [3] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *Hardware Equivalence Checking Problems Submitted to the SAT Competition 2024*. SAT Competition 2024.
- [4] Armin Biere, Tobias Faller, Katalin Fazekas, Mathias Fleury, **Nils Froleys**, and Florian Pollitt. *CaDiCaL, Gimsatul, IsaSAT and Kissat Entering the SAT Competition 2024*. SAT Competition 2024.
- [5] **Nils Froleys**, Emily Yu, and Armin Biere. *ReconfAIGERation entering Core Challenge 2022*. Core Challenge 2022.
- [6] **Nils Froleys**, Emily Yu, and Armin Biere. *Unique Reconfiguration Sequence*. SAT Competition 2022.
- [7] Emily Yu, **Nils Froleys**, Armin Biere, and Mathias Fleury. *Hardware Model Checking Certificates*. SAT Competition 2022.
- [8] Tomás Balyo, **Nils Froleys**, Marijn JH Heule, Markus Iser, Matti Järvisalo, and Martin Suda. *Proceedings of SAT Competition 2020 : Solver and Benchmark Descriptions*. 2020
- [9] **Nils Froleys**, Tomás Balyo, and Dominik Schreiber. *PASAR Entering the Sparkle Planning Challenge 2019*. Sparkle Planning Challenge 2019.